

The slow experiment: searching for self-amplifying handedness at nature’s speed

Aim

Test whether simple early-Earth chemistry in water can **amplify a small excess of one hand** when given months to years instead of days. The spec sheet (README Part 6) shows that speed is never the problem for nature, yet lab searches usually stop after days.

Hypothesis (stated before any data)

At least one prebiotic, water-based peptide-forming system shows **seed-following growth of enantiomeric excess** (ee) at meteoritic concentrations ($\leq 1\text{ mM}$), with an effective rate $k_2 \geq 3 \times 10^{-6}\text{ /M/s}$. That rate is slow enough to be missed by short experiments but fast enough to matter in a 1 mM pond over 100 years (the “pond floor”).

Revised 2026-09-30 after expert feedback. The first version ran at 0.1 M with an assumed 0.2% ee precision. Meteoritic isovaline reaches about 20 nmol/g of rock where it carries an excess (Murchison) and 244 nmol/g where it is racemic (EET 92042), about 0.2–2 mM in parent-body water at a water-to-rock ratio of 0.1. Published ee uncertainties are $\pm 0.01\text{--}1.5\%$ (GC-MS) and 1.2–7.2% (LC-MS), and 2.6% is the standard error of 20 measurements of Murchison isovaline (Glavin & Dworkin 2009; their racemic standard read $-2.3 \pm 1.3\%$, $n = 14$).

Why a *seeded* design

A reaction that only copies itself keeps ee constant, because both hands grow equally. ee *grows* only if the hands also **suppress each other**. So watching ee grow from a small seed tests self-copying and mutual suppression together (spec R1 + R2), which is exactly the missing combination. Seeding is also far faster than waiting for symmetry to break from pure 50/50. From 50/50, noise in a 50 μL , 1 M vial would need about 18 e-foldings of growth to become visible, versus under 1 from a 5% seed.

How long and how concentrated (design.py, review.py R5)

Smallest detectable rate (/M/s) with a **20% seed**, similar to the largest isovaline excesses in Murchison. Each time point is measured 9 times, so the noise is $\sigma/3$:

concentration	ee noise per measurement	1 month	6 months	1 year	3 years
1 mM (realistic)	2.6%	6×10^{-5}	1×10^{-5}	5×10^{-6}	2×10^{-6} ✓ pond
1 mM	1%	3×10^{-5}	4×10^{-6}	2×10^{-6} ✓	7×10^{-7}
10 mM (accelerated)	2.6%	6×10^{-6}	1×10^{-6}	5×10^{-7}	2×10^{-7}
0.1 M (accelerated)	2.6%	6×10^{-7}	1×10^{-7}	5×10^{-8}	2×10^{-8}

Nature’s floors at realistic concentrations: **pond 3×10^{-6}** (1 mM, 100 yr); **ocean 3×10^{-9}** (1 μ M, 10^8 yr). No lab run can reach the ocean floor at natural concentration.



Choice: two tiers. 1. **Realistic arm:** 1 mM, 20% seed, 3 years, 9 independent vials per condition. Reaches 1.8×10^{-6} , below the pond floor (3 vials would reach 2.9×10^{-6} , only 8% below it). This asks directly whether amplification happens at meteoritic concentrations. 2. **Accelerated arm:** 10 mM and 0.1 M for 1 year. These are **not natural conditions**. The arm exists to find any amplifier at all and to measure how the rate of ee growth scales with concentration (the rate law). The model needs that scaling to extrapolate to 1 μ M. If ee grows faster at 0.1 M than at 10 mM by exactly 10 $\times$, k_2 is a true second-order constant and the extrapolation is fair.

Arms (chemistries to test)

Chosen from the scorecard for prebiotic inputs (R4) and biological products (R7):

arm	system	why	source
A	amino acids + carbonyl sulfide (volcanic gas) $\rightarrow$ peptides, in water	prebiotic peptide formation; peptides can prefer same-hand partners	Leman, Orgel & Ghadiri 2004
B	amino acids + hydroxy acids, wet-dry cycling (e.g. daily)	pond-like driving; makes polypeptides	Forsythe et al. 2015
C	prebiotic cysteine peptides that catalyse peptide joining in neutral water	the closest known thing to self-copying in prebiotic water	Foden et al. 2020
D	RNA precursor (RAO) on magnetite , then crystallization	non-autocatalytic route that already breaks symmetry	Ozturk et al. 2023
E	peptide-catalysed transamination (pyruvate + pyridoxamine $\rightarrow$ alanine) coupled to peptide ligation , run fed (slow flow)	a proposed prebiotic network predicted (in a model) to break symmetry on its own, built from two reactions studied in the lab; the model says it must be fed, not closed (review.py R6)	Yu et al. 2024; Deng, Yu & Blackmond 2024; Higgs & Blackmond 2025
+	Positive control: Viedma grinding of a known conglomerate	proves the pipeline detects real amplification	Viedma 2005

Arms E and C are the priority. E is a prebiotic network with a published model prediction that it may break symmetry on its own (Higgs & Blackmond 2025). Its transamination step has been run in the lab but its rate is not yet pinned down: Yu et al. 2024 Table 2 reports % conversions for the *reverse* reaction (alanine + pyridoxal $\rightarrow$ pyruvate + pyridoxamine), which imply rough second-order rates of 6×10^{-6} – 1×10^{-4} /M/s depending on the

peptide catalyst (our estimate, not a measured rate constant). That is a conversion rate, not the amplification rate k_2 in the table above, so it does not show that E's amplification would be detectable; measuring k_2 is the point of the arm. (Corrected 2026-10-01: earlier text called E "the only" such network and said its step "has a measured rate near 10^{-4} /M/s ... well above the 1-year detection floor"; the 10^{-4} came from the reverse reaction and the floor applies to k_2 , not to conversion.) Run E in a slowly fed vessel; a closed vial is predicted to return to 50/50. C is the only other catalytic loop in prebiotic water.

Conditions per arm

For each arm, three seeds $\times$ 9 independent vials (realistic arm; 3 suffice for the accelerated arm): - **+20% L seed** - **+20% D seed** (the mirror control, the most important control) - **0% (racemic)**

Plus three negative controls (3 vials each): no activator / no cycling; sterile-filtered; and the seed amino acid alone (to measure plain racemization). At every time point, also run a procedural blank: no amino acid, carried through the same sampling, workup and analysis as the real vials. It should show no amino acids; any L or D it shows is contamination from handling or reagents and sets the floor for that time point. That's $27 + 9 = 36$ vials per realistic arm ($9 + 9 = 18$ per accelerated arm), plus the positive control. Temperature 25 °C, with an optional 40 °C set to speed things up. Racemization is then negligible over a year for most amino acids, and the "seed alone" vials measure it directly.

Sampling

Days 0, 3, 7, 14, 30, 60, 90, 120, 180, 270, 365, then every 6 months to 3 years for the realistic arm (roughly evenly spaced on a log scale). Each vial is injected in triplicate; injections are averaged, not counted as replicates. Measure ee of the monomers (and of short peptides where possible) by **chiral GC-MS or HPLC** on derivatized samples. Keep analysts **blind** to vial labels.

Decision rule (fixed in advance)

Call an arm an **amplifier** only if **all four** hold: 1. |ee| rises by $> 3.7\%$ absolute over its start in both seeded sets ($3\sqrt{2} \times 2.6\%/\sqrt{9}$, taking 2.6% per vial). Use a smaller threshold if the lab's measured precision is better. 2. The **L-seeded and D-seeded vials change by equal and opposite amounts** (within noise). 3. The racemic vials stay at 0 within noise. 4. Sterile and no-activator controls show no change, and the procedural blanks show no amino acids.

Then fit $ee(t) = ee_0 \cdot \exp(k \cdot t)$ to get k , and $k_2 = k / c$. Compare k_2 with the floors (3×10^{-6} pond at 1 mM, 3×10^{-9} ocean at 1 μ M), using the accelerated arm's rate law to extrapolate.

The eight tests for a credible excess (Dworkin et al. 2024)

Dworkin et al. (2024) list eight tests that a reported ee should pass, and say that any test not passed should be explained. How this plan handles each:

test	what it asks	how this plan meets it
1 Blanks	lab blanks shown	sterile-filtered and no-activator controls, plus a procedural blank (no amino acid) at every time point (Conditions)
2 Isomers	all structural isomers accounted for	defined starting materials, so the possible isomers are known; run standards of every enantiomer the chemistry can make (including peptides measured as monomers)
3 Co-elution	no unrelated compound under the peak	confirm each peak by mass spectrum, and on a second column or derivatization; the racemic vials show where a peak sits when nothing has changed
4 Isotopes	$\delta^{13}\text{C}$ of each enantiomer, separately	does not apply in the usual form: there is no terrestrial-vs-extraterrestrial question in a lab synthesis. Optional: a ^{13}C -labelled seed in one replicate set would show whether new ee comes from the seed or from the feedstock
5 Errors	statistics from separate replicates, not repeat integrations of one run	resolved: the realistic arm uses 9 independent vials per seed condition, so the noise is $\sigma/3$ and the arm reaches 1.8×10^{-6} (review.py R5). With 3 vials it would reach 2.9×10^{-6} , only 8% below the pond floor. Repeat injections of one vial do not capture vial-to-vial scatter and are averaged, not counted. Report the number of vials and of injections separately
6 History	contamination sources and sample history	biology adds L, so the D-seeded mirror arm is the key check; sealed vials, a chain-of-custody log, and blind analysts
7 Context	ee consistent with the rest of the sample	an excess must come with matching chemistry: products form, the yield is sensible, and the excess appears only in the arms that make them; the racemic set staying at 0 is the internal positive and negative check
8 Multiple examples	same result in several samples or labs	nine independent vials per seed condition, the mirror arm, and the Viedma positive control; any amplifier

test	what it asks	how this plan meets it
		found should be re-run in a second lab before it is claimed

Contamination: the main way this goes wrong

Biology is left-handed, so any microbe, skin trace or reagent impurity adds **L**. That would look like “amplification” in the L-seeded vials. Safeguards: - **The D-seed mirror arm**: real amplification is symmetric; contamination pushes both sets toward L. - Sterile filtration, sealed vials, and a vial with no amino acid (it should show zero amino acids). - If only the L arm “amplifies”, report it as contamination, not a result.

What each outcome means

result	meaning for the project
An arm amplifies at $k_2 \geq$ pond floor	First prebiotic, water-based amplifier; pond seeding by meteorites becomes a complete story
An arm amplifies but slower than the floor	Real, but too slow to matter in a pond; ocean-scale only
Nothing amplifies at 1 mM in 3 years, or at 0.1 M in 1 year	Rules out these chemistries down to $k_2 \approx 2 \times 10^{-6}$ (realistic) or 5×10^{-8} (accelerated). That’s still a publishable upper limit, and it tightens the spec
Only the positive control amplifies	Pipeline works; the gap in the scorecard is real

Limits and practical notes

- This needs a chemistry lab with chiral GC-MS or HPLC. It’s a proposal for a mentor or a summer lab, not a home experiment. Local groups with chiral GC-MS: NASA Ames (G. Cooper, G. Chaban); San José State (A. Rios). Carbonyl sulfide is a toxic gas and needs fume-hood handling (arm A).
- The 2.6% noise figure is a published standard error (20 measurements), used here as an assumed per-vial error; a single measurement scatters more. The lab’s own precision sets the real threshold; the smallest detectable k_2 grows roughly in proportion to the noise.
- A null result limits only these specific chemistries, not all possible ones.
- The rate model is the single-exponential early phase from `sim.py`. Late saturation isn’t used for the fit.

References

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